Experiment 35

Aim

To perform morphological Black Hat operation on an image using OpenCV.

Procedure

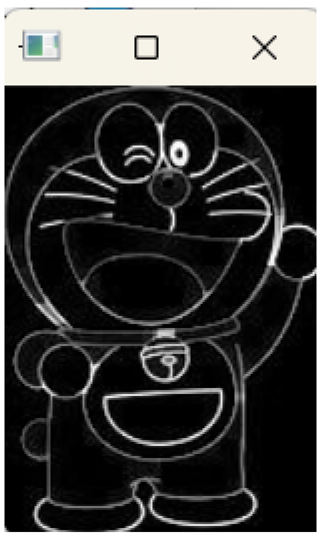
1. Import the required libraries, OpenCV and NumPy.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a morphological kernel.
5. Apply the Black Hat operation using cv2.morphologyEx().
6. Display the original and Black Hat images.
7. Save the resulting image.

Code:

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found!")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.ones((5, 5), np.uint8)
black_hat = cv2.morphologyEx(gray, cv2.MORPH_BLACKHAT, kernel)
cv2.imshow("Original Image", img)
cv2.imshow("Black Hat Image", black_hat)
cv2.imwrite("black_hat_image.jpg", black_hat)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:

Output:



Result

The morphological Black Hat operation was successfully performed using OpenCV. The Black Hat operation highlighted small dark regions and details in the image.